

Lecture 2 (slide 33)

① Sales of new one-family houses, USA

- T: The trend fluctuates around a constant level; not strong.
- S: There is yearly seasonality present, made up of 4 quarters.
- C: Multiple rises and falls indicate cyclicality.
- R: Random variation is present.

Stationarity: The presence of seasonality indicates that the mean is not constant. Variation does not appear to be constant → non-stationary time series.

② US treasury bill contracts (daily observations)

- T: There is a decreasing linear trend.
- S: There is no clear indication of seasonality.
- C: There is no clear indication of cyclicality.
- R: Random variation is present.

Stationarity: The presence of a trend indicates that the mean is not constant. Variation changes slightly over time and is not perfectly constant.

③ Australian quarterly electricity production

- T: There is an increasing non-linear trend.
- S: There is yearly seasonality present, made up of 4 quarters.
- C: There is no indication of cyclicality.
- R: Random variation is present.

Stationarity: The presence of seasonality and a trend indicates that the mean is not constant. Variation diverges over time and is not constant.

④ Google daily closing stock price

- T: The series fluctuates around a constant level.
- S: There is no clear indication of seasonality.
- C: There is no indication of cyclicality.
- R: Random variation is present.

Stationarity: The series fluctuates around a constant mean, but the variance is only broadly stable.

(leaning towards "weakly stationary" — a concept not discussed in this module). For STA2020 purpose: based on variation the series is non-stationary.

Lecture 3 (Slide 54)

add observations: 8.5 4.7 6.0

See slide 66 for memorandum for answers up to 1999

<u>Year</u>	<u>t</u>	<u>y_t</u>	<u>MA(3)</u>	<u>MA(4)</u>	<u>CMA(4)</u>
2000	9	3.5	3.67	4.38	4.19
2001	10	5.5	5.17	6.00	5.19
2002	11	6.5	6.83	6.30	6.15
2003	12	8.5	6.57	6.43	6.36
2004	13	4.7	6.4	X	X
2005	14	6.0	X	X	X
